Introduction to Database Management System

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Database Management System

Definition (Database Management System)

A Database Management System is a collection of interrelated data and set of programs to access those data.

- The collection of data, usually referred to as the database.
- The primary goal of the DBMS is to provide a way to store and retrieve database information that is both convenient and efficient.
- Management of data involves both defining structures for storage of information and providing mechanism for the manipulation of the information.
- The database system must ensure the safety of the information stored, despite system crashes or attempts at unauthorized access.

Database-System Application

- Database systems are used to manage collections of data that:
  - are highly valuable
  - are relatively large, and
  - are accessed by multiple users and applications, often at the same time.
- University Database vs. Social Media Platform Database
- Modern Database exploit commonalities in the structure of data to gain efficiency, but also allow for weakly structured data and for data whose formats are highly variable.
- As a result they are large and complex.

Database-System Application

- By providing high level of abstraction, a database system makes it possible for an enterprise to combine data of various types into a unified repository of the information needed to run enterprise.
  - Enterprise Information
    - Sales
      - Accounting
      - Human Resources
  - Manufacturing
  - Banking and Finance
    - Banking
    - Credit Card and Transactions
    - Finance
  - Universities
  - Airlines
  - Telecommunications
  - Web-based services
    - Social-media
    - online retailers
    - online advertisement
  - Document Database
  - Navigation Systems

Database-System Applications

- The ways in which people interact with databases has changed over time.
- Early database were maintained as back-office systems with which user interacts via print reports and paper forms for input.
- Once the bank customer has to interact with a teller for every transaction, Automated-teller machine (ATM) allows direct user interaction.
- The user, or consumer, can focus on the product or service without being aware of the details of the large database that makes interaction possible.
- There are two modes in which database are used
  - Online transaction processing: Large number of users use the database, with each user retrieving relatively small amounts of data and performing small updates.
  - Data analytics: The processing of data to draw conclusions, and infer rules or decision procedures, which are then used to drive business decisions.

Case-Study: File Processing System

- Consider university organization that, among other data keeps information about all teachers, students, schools and course offering.
- One way to keep information on a computer is to store it in operating-system files.
- User can perform manipulation via numbers of applications provided by the Operating System. Consider that program allows
  - Add new student, teacher and course.
  - Register students for course and generate class rosters.
  - Assign grades to students, compute grade points average, and generate transcripts.
- New Program are added to the system as the need arise.
- This typical file-processing system is supported by conventional Operating System.
- The system stored permanent records in various files, and it needs different application programs to extract records from, and add records to, the appropriate files.

Purpose of Database Systems

- Keeping organizational data in file processing system has number of disadvantages
  - Data Redundancy and inconstancy
    - Different programmers creates the files and application program over a long period.
    - Various files are likely to have different data-structure.
    - Programs may be written in several programming languages
    - Moreover, the same information may be duplicated in several places (files in our case).
    - Redundancy leads to higher storage coast.
    - It may lead to Inconsistency; i.e. various copies of the same data may no longer agree.
  - Difficulty in accessing data
    - Access to the data is limited to the reports, that programs generate.
    - A program can generate report that contains list of students.
    - How can we get list of students that are from same state/area?
    - We can filter manually or Write new program.
    - What if requirement change ? now, i need students with 7.5 CGPA....
  - Data Isolation
    - Because Data is scattered in various files, and files may be in different formats, writing new application program to retrieve the appropriate data is difficult.

Purpose of Database Systems

- Integrity Problems
  - Data values stored in database must satisfy certain types of consistency constraints.(e.g. balance amount must not fall below zero.)
  - Developers can enforce such constraints with code.
  - What if a new constraint needs to be added? and what if that constraint involve several data item from different files.
- Atomicity Problem
  - If failure occurs, the data must be restored to the consistent state that existed prior to the failure.
- Concurrent-access anomalies
  - Concurrent update may lead to inconstant state of the Database.
- Security
  - Not every user should be able to access all the data.

View of Data

A major purpose of a database system is to provide users with and abstract view of the data, i.e. system hides certain details of how the data are stored and maintained.

- Underlying the structure of a database is the data model
- Data model is a collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.
- Data models can be classified into four different categories:
  - Relational Model
  - Entity-Relationship Model
  - Semi-structured Data Model
  - Object-Based Data Model

Relational Model

- It uses collection of tables to represent both data and the relationships among those data.
- Each table has multiple columns and each column has a unique name.
- Tables are also known as relations.
- Relational Model is an example of record-based model.
- The relational model is widely used data model.

Entity-Relationship Model

- The entity-relationship (E-R) data model uses a collection of basic objects called entities, and relationships among these objects.
- An entity is "thing" or "object" in the real world that is distinguishable from other objects.

Semi-structured Data Model

- It permits the specification of data where individual data items of the same type may have different sets of attributes (contrast to the all data models mentioned earlier)
- JSON and XML are widely used semi-structured data representations.

Object-Based Data Model

- Today the concept of objects is well integrated into relational databases.
- Database systems allow procedures to be stored in the database system and executed by the database system.

Data Abstraction

- Physical Level
  - The lowest level of abstraction describes how the data are actually stored.
- Logical Level
  - The next Higher level of abstraction describes what data are stored in the database.
  - Although implementation of the simple structures at the logical level may involved complex physical-level structures, the user of the logical level does not need to be aware of this complexity
- View Level
  - The highest level of abstraction.
  - It describes only part of the database.
  - Even though the logical level uses simpler structures, complexity remains because of the variety of information stored in a large database.
  - At the view level, several view of the database are defined.
  - At the view Level, computer users see a set of application programs that hide details of the data types.

Instances and Schemas

- Database change over a time as information is inserted and deleted.
- The collection of information stored in the database at a particular moment is called an instance of the database.
- The overall design of the database is called the database schema.
- Database system has several schemas, partitioned according to the levels of abstraction
  - Physical Schema: Describes the database design at the physical level.
  - Logical Schema: Describes the database design at the logical level.
  - A database may also have several schemas at the view level, sometimes called subschemas.
  - Each subschema describe the different view of the database.

Database Languages

- Data-definition Language (DDL)
  - We specify database schema using DDL
  - A special type of DDL called data storage and definition language specifies storage structure and access methods.
  - DDL provides facilities to specify constraints (like, account balance must never be negative).
- Data-manipulation Language (DML)
  - it enables users to access or manipulate data as organized by appropriate data model.
  - The types of access are:
    - Retrieval of information.
    - Insertion of new information.
    - Deletion of information.
    - Modification of information stored.

Data-Definition Language

- The DDl provides facilities to specify Constraints.
- A constraint can be an arbitrary predicate pertaining to the database that are costly to test.
- Thus, Database System implement only those integrity constraints that can be tested with minimal overhead:
  - Domain Constraints:
    - A domain of possible values must be associated with every attribute(e.g. integer types, character types, date/time types)
    - Declaring an attribute to be of particular domain acts as a constraint on the values that it can take.
  - Referential Integrity
    - There are some cases where we want to ensure that a value that appears in one relation for a given set of attributes also appears in certain set of attributes in another relation.
    - A database modification can cause violation of referential Integrity, and it is a normal procedure to reject such actions.

Data-Definition Language

- Integrity Constraints contd...
  - Authorization
    - ★ We may want to differentiate among the users as far as the types of access they are permitted on various data values in the database.
    - ★ Most common authorizations are read authorization, insert authorization, update authorization and delete authorization.
    - ★ user may be assigned all, none or combination of these types of authorization.
- Processing of DLL generates some output, that is placed in data dictionary.
- The data dictionary contains Metadata.
- The data dictionary is considered as a special type of table that can be accessed and update by the database system itself
- The database system consults the data dictionary before reading or modifying actual data.

Data-Manipulation Language

- There are basically two types of data-manipulation language:
  - Procedural DMLs: require a user to specify what data are needed and how to get those data
  - Declarative DMLs (Nonprocedural DMLs): require a user to specify what data are needed without specifying how to get those data.
- query is a statement requesting the retrieval of information.
- the portion of a DML that involves information retrieval is called query language.

Database Design

- Database design mainly involves the design of the database schema.
- The initial phase of database design, is to characterize fully the data needs of the prospective database users. The database designer needs to interact extensively with domain experts and users to carry out this task.
- Next, the designer chooses a data model, and by applying the concepts of the chosen data model, translates the requirements into a conceptual schema of the database.
- In terms of the relational model, the conceptual-design process involves decisions on what attributes we want to capture in the database and how to group these attributes to form the various tables
- The process of moving from an abstract data model to the implementation of the database proceeds in two final design phases
  - In the logical-design phase, the designer maps the high-level conceptual schema onto the implementation data model of the database system that will be used.
  - The designer uses the resulting system-specific database schema in the subsequent physical-design phase, in which the physical features of the database are specified.

Database Engine

- The functional components of a database system can be broadly divided into
  - Storage manager
  - query processor
  - transaction management

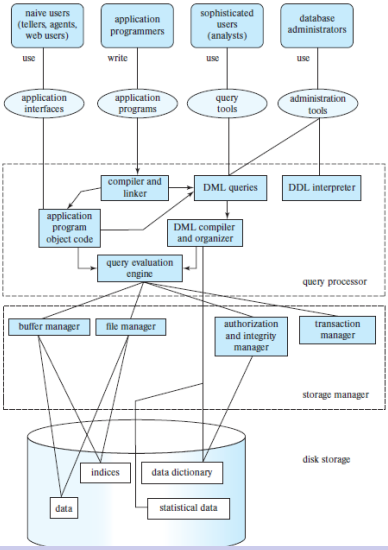
Storage Manager

- The storage manager is the component of a database system that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
- translates DML statements into low-level file-system commands.
- Storage manager components include:
  - Authorization and integrity manager
  - Transaction manager
  - File manager
  - Buffer manager
- The storage manager implements several data structure as part of the physical system implementation
  - Data files
  - Data Dictionary
  - Indices

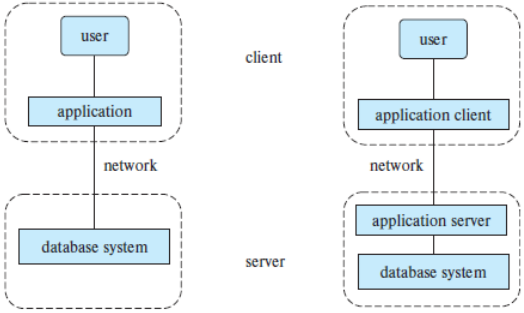
The Query Processor

- DDL interpreter: interprets DDL statements and records into the data dictionary
- DML compiler: translate high-level DML statements in a query into low-level instructions that query-evaluation engine understands
- Query evaluation engine: executes low-level instructions generated by the DML compiler

Database system architecture



Database Application Architecture



(a) Two-tier architecture (b) Three-tier architecture

Database User and User Interfaces

- Naive users
  - interact with the system by using predefined user interfaces, such as web or mobile applications.
  - The typical user interface for naive users is a forms interface, where the user can fill in appropriate fields of the form.
  - Naive users may also view read reports generated from the database.
- Application programmers
  - computer professionals who write application programs.
  - Application programmers can choose from many tools to develop user interfaces.
- Sophisticated users
  - interact with the system without writing programs.
  - form their requests either using a database query language or by using tools such as data analysis software.

Database Administrator (DBA)

- Schema Definition through DDL
- Storage Structure and access-method definition
- Schema and physical organization modification
- Grant authorization for data access
- Routine maintenance
  - Periodic backup
  - Ensure that there is enough space
  - Monitoring jobs running on the database